

Research in Information Security

Dr. Ashok Kumar Das

IEEE Senior Member

Fellow (IET), UK; Fellow (IETE); Fellow (IE)

Web of Science (Clarivate™) Highly Cited Researcher

Professor

Center for Security, Theory and Algorithmic Research
International Institute of Information Technology, Hyderabad
(Formerly, Indian Institute of Information Technology, Hyderabad)

E-mail: *ashok.das@iiit.ac.in*

Homepage: <https://www.iiit.ac.in/faculty/ashok-kumar-das/>
<https://sites.google.com/view/iitkgpdas/>

Welcome to Research in Information Security

Course Instructors

- **Dr. Ashok Kumar Das** (e-mail: ashok.das@iiit.ac.in) [PhD, IIT Kharagpur]
- **Dr. Kannan Srinathan** (e-mail: srinathan@iiit.ac.in) [PhD, IIT Madras]

Why Research?

- **“If we knew what it was we were doing, it would not be called research, would it?” – Albert Einstein**
- **“If I had an hour to solve a problem I’d spend 55 minutes thinking about the problem and 5 minutes thinking about solutions.” – Albert Einstein**
- **“Successful people are not gifted; they just work hard, then succeed on purpose.” – G. K. Nielson**
- **“The three great essentials to achieve anything worthwhile are, first, hard work; second, stick-to-itiveness; third, common sense.” – Thomas A. Edison**

What is Research?

- Research means to carefully analyze the problems or to do the detailed study of the specific problems, by making use of special scientific methods.
- Research can be done on any topic, be it medical, non- medical, Information Technology (IT), or anything else.
- In order to do research, first of all, you need to have a topic or the problem on which you can do research. The topic must have relevant questions to answer.

(**Formulation of Problem Statement**)

- For research, certain steps have to be followed like first observation, then background research then preparing of hypothesis, eventually conducting a simple experiment.

(**Finding Solutions to the Defined Problems**)

- Working on a research project will obviously be a challenging and rewarding experience, provided you put the best of your expertise and skill in it.
- It is an opportunity which helps you to pursue an in-depth or deep original study about any topic which interests you.
- The main aim of the goals is to provide the best of the solution to some of the world problems and also to enhance our knowledge.

Research Paper Structure

Sections	What to do?
Abstract	Write 150 words on the purpose of the study, research questions or suggestions, and main findings with conclusions.
Outline	Organize the map of the study.
Introduction	Provide the main information on the problem statement, the indication of methodology, important findings, and principal conclusion.
Literature Review	Analyze and incorporate scholarly sources on past studies.
Methodology or Materials and Methods	Explain the design of the research with techniques that are used for gathering information and other aspects related to the experiment.
Results	Present and illustrate the obtained findings.
Discussion	Review the information in the introduction part, evaluate their gained results, or compare it with past studies.
Recommendations	Propose potential solutions or new ideas based on the obtained results.
Limitations	Consider the weaknesses of the research and results to get new directions.
Conclusion	Provide final thoughts and the summary of the whole work.
Acknowledgments or Appendix	Include additional information on the research paper if it is necessary.
References	Provide and cite all used sources in the study.

How to Measure Impact of Research?

- The **Science Citation Index (SCI)** is a citation index originally produced by the Institute for Scientific Information and created by Eugene Garfield. It was officially launched in 1964. It is now owned by Clarivate Analytics.
- **Impact Factor:** The impact factor (IF) is a measure of the frequency in which the average article in a journal is cited in a particular year. Impact factors measure the impact of a journal, not the impact of individual articles.
- The **h-index** is an index to quantify an individual's scientific research output. The h-index is an index that attempts to measure both the scientific productivity and the apparent scientific impact of a scientist. The index is based on the set of the researcher's most cited papers and the number of citations that they have received in other people's publications. **h-index is the largest number h such that h publications have at least h citations.**
- The **i10-index** is the number of publications with at least 10 citations.

How to Measure Impact of Research?

- **DBLP:** <https://dblp.org/pid/39/871.html>
- **Google Scholar:** <https://scholar.google.com/citations?user=bToAUHMAAAAJ&hl=en>
- **Web of Science (Clarivate™):** <https://www.webofscience.com/wos/author/record/U-2790-2019>
- **Scopus:** <https://www.scopus.com/authid/detail.uri?authorId=55450732800>
- **My more details on research publications:** <https://sites.google.com/view/iitkgpkdas/research-publications>

- Digital Signatures and Its Variants: Proxy Signature, Multi-Signature and Aggregate Signature, and their applications in Internet of Things (IoT)-enabled environments
- Key management in hierarchical access control
- Key management, user authentication and access control in wireless sensor networks/IoT
- Security in Internet of Vehicles (IoV)
- Intrusion detection
- Blockchain and Its Security and Privacy Issues
- AI/ML Security
- Post-Quantum Cryptographic Protocols
- Multiparty computation (MPC): It allows for multiple parties to share data for computing tasks without revealing each other's data
- Quantum Cryptography, QKD, Quantum Teleportation (QT)

- Books

- ▶ William Stallings, "Cryptography and Network Security: Principles and Practices," Pearson Education, 6th Edition.
- ▶ Bernard Menezes, "Network Security and Cryptography," Cengage Learning.
- ▶ Behrouz A. Forouzan, "Cryptography and Network Security," Special Indian Edition.

- Research papers

- ▶ IEEE Transactions
- ▶ ACM Transactions
- ▶ Elsevier
- ▶ Springer
- ▶ Wiley

Grading Method:: Relative

- Mid Sem: 20% (Closed books and notes exam)
- End Sem: 40% (Closed books and notes exam)
- Project: 40% (Maximum of 3 people group is allowed) [2 Presentations (30%) + Project Report (10%)]

About Instructor

Dr. Ashok Kumar Das

About me: Dr. Ashok Kumar Das

- **Education:** Ph.D. in Computer Science and Engineering, M.Tech. in Computer Science and M.Sc. in Mathematics and Computing from **IIT Kharagpur**, India
- **Visiting Research Professor** at the **Old Dominion University (ODU), Virginia, USA**, with the Virginia Modeling, Analysis, and Simulation Center (2022, 2024)
- **Distinguished Adjunct Professor** at the **Korea University, Seoul, South Korea** from July 2024 onwards [2026 QS World Rank: 61]
- **Research Interests:** Network and system security, blockchain, security in Internet of Things (IoT), Internet of Vehicles (IoV), Internet of Drones (IoD), cloud/fog computing, intrusion detection, AI/ML security, quantum and post-quantum cryptography
- **Research Highlights [Research Publications (Total: 572)]:**
 - ▶ Number of Journal Papers: 493
 - ▶ Number of Conference Papers: 66
 - ▶ Number of Book Chapters: 10
 - ▶ Number of Edited Books/Volumes: 3
 - ▶ Number of IEEE Transactions/IEEE Journal/IEEE Magazine Papers: 243

- **Total citations: 33480, h-index: 102, i10-index: 390**
(According to Google Scholar Citations as on July 27, 2026)
- **Published in top venues like**
 - * IEEE Transactions on Information Forensics and Security
 - * IEEE Transactions on Dependable and Secure Computing
 - * IEEE Transactions on Consumer Electronics
 - * IEEE Transactions on Smart Grid
 - * IEEE Transactions on Industrial Informatics
 - * IEEE Transactions on Vehicular Technology
 - * IEEE Internet of Things Journal
 - * IEEE Consumer Electronics Magazine
 - * IEEE Communications Magazine
 - * IEEE Journal of Biomedical and Health Informatics
 - * IEEE Transactions on Network Science and Engineering
 - * IEEE Transactions on Intelligent Transportation Systems
 - * IEEE Sensors Journal
 - * IEEE Transactions on Network and Service Management
 - * IEEE Transactions on Cloud Computing
 - * IEEE Communications Surveys and Tutorials
 - * ACM Computing Surveys
 - * ACM Transactions on Sensor Networks

About me: Journal Editorial Board Members

- Associate Editor: **IEEE Transactions on Information Forensics and Security** (SCI Impact Factor: 6.8)
- Associate Editor: **IEEE Systems Journal** (SCI Impact Factor: 4.4)
- Editor: **Journal of Network and Computer Applications (Elsevier)** (SCI Impact Factor: 8.7)
- Editor (Technical Committee): **Computer Communications (Elsevier)** journal (SCI Impact Factor: 6)
- Associate Editor: **Journal of Cloud Computing (Springer)** (SCI Impact Factor: 4)
- Associate Editor: **Alexandria Engineering Journal (Elsevier)** (SCI Impact Factor: 6.8)
- Associate Editor: **IET Communications** journal (SCI Impact Factor: 1.6)
- Associate Editor: **Cyber Security and Applications (Elsevier)**
- Editor: **KSII Transactions on Internet and Information Systems** (SCI Indexed Journal)
- Editor: **International Journal of Communication Systems (Wiley)** (2024 -) (SCI Impact Factor: 1.7)
- Guest Editor: **IEEE Journal of Biomedical and Health Informatics**, 2026
- Guest Editor: **IEEE Transactions on Consumer Electronics**, 2025
- Guest Editor: **Ad Hoc Networks (Elsevier)**, 2026
- Guest Editor: **Computers & Electrical Engineering (Elsevier)** (SCI Indexed Journal), 2016
- Guest Editor: **ICT Express (Elsevier)** (SCI Indexed Journal), 2019
- Guest Editor: **Wireless Communications and Mobile Computing** (SCI Indexed Journal), 2020

About me

- Panel member at the prestigious **World Laureates Summit (WLS 2026)**, held in Dubai, UAE, from 1–3 February 2026, jointly organized with the World Government Summit (WGS 2026), on “Blockchain × Science Forum — Science in the Web3 Era: Foundations and Opportunities.”
- Mentor – IIM Lucknow Enterprise Incubation Centre (CoE–Blockchain Technology), supported by the Government of Uttar Pradesh, India, under the StartInUP Policy and IIML-EIC onboarded Microsoft as the Technology Partner for CoE-BT, 2026.
- Member of Panel Discussion: Security by Design: Embedding Trust, Ethics, and Transparency into AI Systems, at Cybersecurity and Privacy Workshop, 18th IEEE International Conference on COMMunication Systems & NETworks (COMSNETS 2026), January 6 - 10, 2026, Bangalore, India.
- Member of the Technical Program Committess (TPCs) for **MAHA-AI for Science & Engineering (AI-SE) Mission of Anusandhan National Research Foundation (ANRF), Department of Science & Technology**, New Delhi, Government of India.
- Visiting Research Professor at the **Old Dominion University (ODU)**, Suffolk, VA 23435, USA, with the Virginia Modeling, Analysis, and Simulation Center (VMASC) [2022, 2024]
- **Distinguished Adjunct Professor** at the **Korea University, Seoul, South Korea** [2025 QS World Rank: 67], for two-year term starting from July 2024
- Listed in the **Web of Science (Clarivate™) Highly Cited Researcher 2022, 2023** in recognition of exceptional research performance demonstrated by production of multiple highly cited papers that rank in the top 1% for field and year.
- 2025–2022 - Research.com Computer Science in India Leader Awards

- **Mauro Conti**, IEEE Fellow, Head of SPRITZ Security and Privacy Research Group, Director of UniPD node of CINI Cybersecurity National Lab, EU Marie Curie Fellow Alumni, CEO and co-funder of CHISITO, and Co-funder of DYALOGHI, University of Padua, Italy
- **Willy Susilo**, IEEE Fellow, ARC Future Fellow, Co-Director, Centre for Computer and Information Security Research, University of Wollongong, Australia
- **Sajal K. Das**, IEEE Fellow, Professor and Daniel St. Clair Endowed Chair, Department of Computer Science, Missouri University of Science and Technology, Rolla, MO 65409, USA
- **Sherali Zeadally**, Fellow of the British Computer Society and the Institution of Engineering Technology, Stevenage, U.K., University of Kentucky, Lexington, KY 405 06 USA
- **Kim-Kwang Raymond Choo**, Fellow, Australian Computer Society, Department of Information Systems and Cyber Security, The University of Texas at San Antonio, San Antonio, TX 78249 USA
- **Xinyi Huang**, Fujian Provincial Key Laboratory of Network Security and Cryptology, Fujian Normal University, Fuzhou, Fujian, China
- **Alexey Vinel**, Halmstad University, Halmstad, Sweden
- **Muhammad Khurram Khan**, FIET (UK), FBCS (UK), Center of Excellence in Information Assurance, King Saud University, Riyadh, Saudi Arabia
- **Bharat Bhargava**, Purdue University, West Lafayette, Indiana, USA

- **Athanasios V. Vasilakos**, Lulea University of Technology, Sweden
- **Minho Jo**, Department of Computer and Information Science, Korea University, Seoul, South Korea
- **Laurence T. Yang**, St. Francis Xavier University, Canada
- **Joel J. P. C. Rodrigues**, IEEE Fellow, National Institute of Telecommunications - Inatel, Brazil
- **Debiao He**, School of Cyber Science and Engineering, Wuhan University, Wuhan 430 072, China
- **Jong-Hyouk Lee**, Sangmyung University, Republic of Korea
- **Kee-Young Yoo**, Kyungpook National University, Daegu, Korea
- **Qi Jiang**, Xidian University, China
- **Sachin Shetty**, Old Dominion University, USA
- **Xiong Li**, Hunan University of Science and Technology, China
- **Mamoun Alazab**, Charles Darwin University, Australia
- **Mohammad S. Obaidat**, IEEE Fellow, University of Sharjah
- **YoungHo Park**, School of Electronics Engineering, Kyungpook National University, South Korea
- **Shantanu Pal**, School of Information Technology, Deakin University, Melbourne, Australia

Thank you!
Questions!